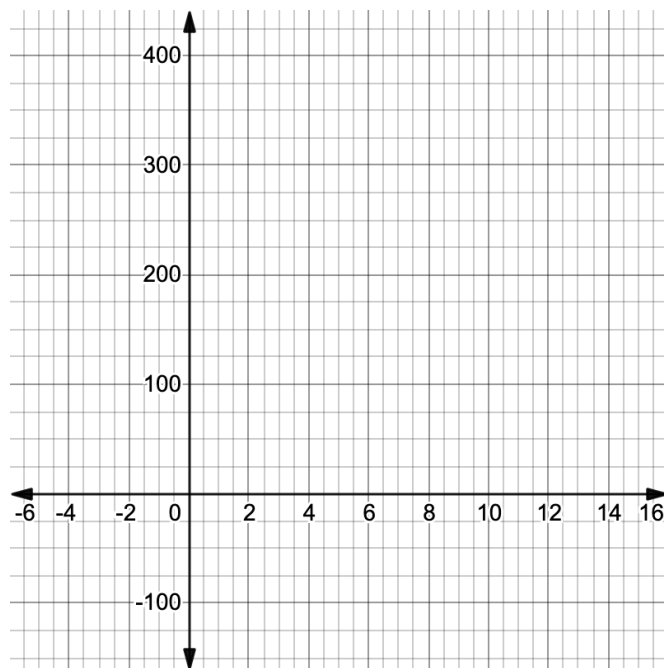




4.7.4 Wrap-Up: Ticket Out the Door

1. A Florida Department of Forestry scientist has been tracking the population of the Monarch Butterfly since the 2005 migration cycle. The scientist uses the function $B(x) = -35x + 325$ to estimate the number of butterflies, where x is the number of years since 2005 and B is the number of butterflies, in thousands.

- a. Graph the function in terms of the context.



- b. Find the solution to $B(x) = 0$.
- c. Determine if the solution to $B(x) = 0$ is viable. Explain your answer.



Unit 4, Lesson 8: Comparing Linear Functions



Benchmark

MA.912.F.1.5 Compare key features of linear functions each represented algebraically, graphically, in tables, or written descriptions.



Learning Target

- ☐ I can compare key features of linear functions.



4.8.1 Warm-Up: Blood Donation

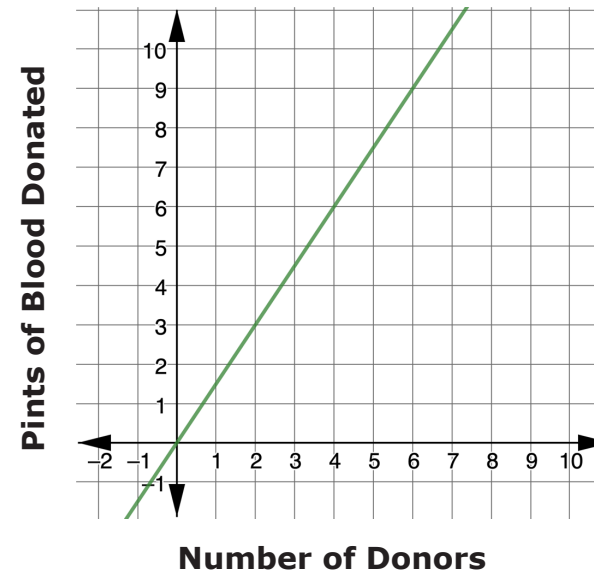
Dr. Charles Drew, a pioneering African American medical researcher, discovered a way to store and process blood for transfusions. Dr. Drew managed two of the largest blood banks in the world during World War II as the medical director of the Red



Cross National Blood Donor Service, collecting tens of thousands of pints of blood for U.S. soldiers. It has been said that Dr. Drew's research and work saved the world against Nazism, as battlefield blood storage and transfusions did not exist before.

After a ruling stated that the blood from African-Americans could not be transfused to white soldiers, Dr. Drew resigned from the Red Cross and became a professor at Howard University's medical school (an HBCU), training a generation of black physicians on the research of storing and collecting blood.

1. The graph shows the relationship between pints of blood collected and the number of donors according to the Red Cross in the U.S. today.

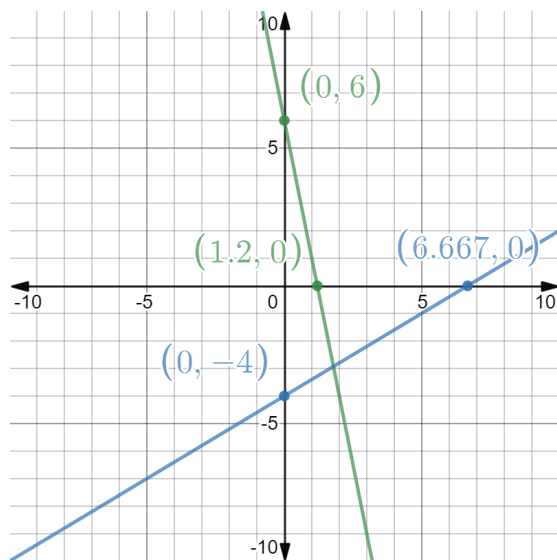


- a. What is the slope of the line?
- b. What does the slope represent?
- c. Write an equation to represent the linear relationship between the number of donors, d , and the pints of blood collected, b .
- d. Use the equation to find the number of pints of blood collected if 12,000 people donate blood.



4.8.2 Collaboration: Comparing Representations of Linear Functions

1. Two linear functions are shown on the coordinate grid.



- Which line has the larger x -intercept?
- Which line has the smaller y -intercept?
- Which line has the larger slope?
- For which line does y decrease as x increases?

2. The table shows other ways the linear relationships could be represented.

	Function	Table of Values		Written Description
Green Line	$f(x) = 1 - 5(x - 1)$	x	y	A line with a slope of -5 that passes through the point $(1, 1)$.
		-2	16	
		1.2	0	
		2	-4	
		7	-29	
Blue Line	$g(x) = \frac{3}{5}x - 4$	x	y	A line that passes through the points $(0, -4)$ and $(6\frac{2}{3}, 0)$.
		-5	-7	
		-1	-4.6	
		0	-4	
		3	-2.2	

- With your partner, discuss which representation(s) of the green line can be used to determine the x -intercept.
- With your partner, discuss which representation(s) of the blue line can be used to determine the y -intercept.

3. Brenda and Austin were discussing the representations of the blue and green line.

Brenda	Austin
I can determine whether the linear relationship is increasing or decreasing as x increases using the slope of the line. If the representation does not have the slope, I have to graph the relationship.	I can determine whether the linear relationship is increasing or decreasing as x increases using the graph, slope-intercept form, point-slope form, or the table.

- a. With your partner, discuss how Brenda can determine if a linear relationship is increasing or decreasing using only the slope of the line. Write a summary of your discussion.

- b. With your partner, discuss how Austin could use a table to determine if a linear relationship is increasing or decreasing. Write a summary of your discussion.

- c. Brenda told her teacher that she could not determine if the blue line is increasing or decreasing from its written description because it did not include the slope. Explain to Brenda how she could use the written description to find the slope.

- d. Work with your partner to help Brenda and Austin understand how to use an equation in standard form, such as the equation that represents the blue line, $3x - 5y = 20$, to determine if the relationship is increasing or decreasing.



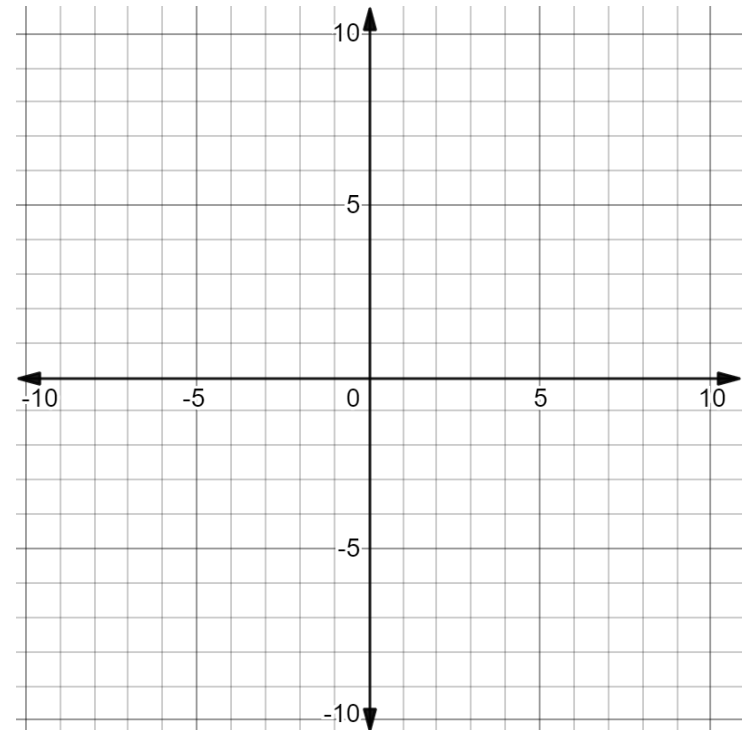
4.8.3 Guided Instruction: Comparing End Behavior

Two different linear functions are shown.

Line A		Line B
x	y	$f(x) = -\frac{4}{9}x + 2$
-18	4	
-12	$\frac{4}{3}$	
0	-4	
2.25	-5	
18	-12	

1. Which linear relationship has the smaller y -intercept?
2. Compare the domains of the two linear relationships.

3. Determine which linear relationship has the larger slope.
4. Determine if line A is increasing or decreasing as x increases. Show your work or explain your thinking.
5. Determine if line B is increasing or decreasing as x increases. Show your work or explain your thinking.
6. Sketch and label the graph of each linear relationship on the coordinate grid.



The **end behavior** of a function refers to the appearance of the function's graph at the "ends," in other words, as the x -values get infinitely smaller or as the values get infinitely larger.

7. Use the definition of end behavior to complete the sentences.

a. For line A, as the x -values get smaller, the appearance of the graph of A approaches

- ☐ $-\infty$.
☐ ∞ .

b. For line A, as the x -values get larger, the appearance of the graph of A approaches

- ☐ $-\infty$.
☐ ∞ .

Another way to represent end behavior is to use mathematical notation. Two examples are shown in the table.

Notation	Meaning
$x \rightarrow -\infty$	" x goes to negative infinity"
$y \rightarrow \infty$	" y goes to infinity"

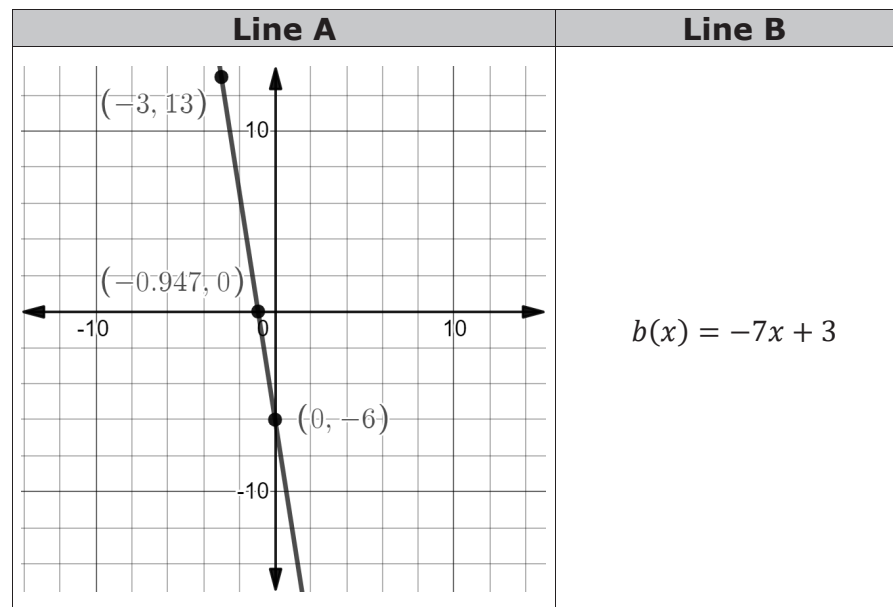
8. Complete the mathematical notation for end behavior for line B.

As $x \rightarrow -\infty$, $y \rightarrow$ _____. As $x \rightarrow \infty$, $y \rightarrow$ _____.



4.8.4 Your Turn!

1. Two different linear functions are shown.



a. Name the key feature(s) that are the same for both linear functions.

b. Give the key feature(s) that are different for the functions.

c. Which linear function has the smaller slope?

d. Which linear function has the larger x -intercept?



4.8.5 Wrap-Up: Reflection

- Complete the table to describe your understanding of each key feature.

Domain	Range
☐ I need help. ☐ I got it. ☐ I could help someone.	☐ I need help. ☐ I got it. ☐ I could help someone.
Slope	x -intercept
☐ I need help. ☐ I got it. ☐ I could help someone.	☐ I need help. ☐ I got it. ☐ I could help someone.
End Behavior	y -intercept
☐ I need help. ☐ I got it. ☐ I could help someone.	☐ I need help. ☐ I got it. ☐ I could help someone.